Notes and Solutions for Semiparametric Theory and Missing Data

A companion to Tsiatis (2006)

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Table of contents

Home

This book develops explanatory notes, derivations, worked examples, and original solutions for exercises accompanying *Semiparametric Theory and Missing Data* by Anastasios A. Tsiatis.

The objective is to make the geometry of semiparametric models and the theory of missing and coarsened data easier to learn and apply. The presentation follows the organization of the source book while remaining independently written.

How to use the book

Each chapter contains the following components:

- Learning objectives and prerequisites
- A notation guide
- Expository notes and geometric intuition
- Expanded derivations
- Worked examples
- Solutions to selected exercises
- Additional exercises and computational illustrations
- A short chapter summary

Source text

Readers should consult Tsiatis (2006) for the original statements, examples, and exercises. Exercise numbers in this companion refer to that text.

Preface

Purpose

These notes are intended for graduate students and researchers in biostatistics and statistics who want a detailed path through semiparametric theory, influence functions, missing-data methods, and causal inference.

Conventions

- Random variables are written in uppercase and realizations in lowercase
- Vectors are treated as column vectors unless stated otherwise
- Expectations and inner products are taken under the true data-generating distribution unless annotated
- A solution labeled "Exercise *j.k*" corresponds to that numbered exercise in Tsiatis (2006)

Scope and attribution

This is a companion volume, not a reproduction of the source. Explanations, derivations, examples, and solutions should be written independently. Page and exercise references identify where the corresponding topic appears in the source.

1 Notation

Symbol	Meaning
Z	Full observed random vector for one independent unit
$Z_1, \dots, Z_n$	Independent and identically distributed observations
β	Finite-dimensional parameter of interest
η	Nuisance parameter, often infinite-dimensional
S_β	Score for β
Λ	Tangent space
$\Lambda^\perp$	Orthogonal complement of a tangent space
φ	Influence function
φ_{eff}	Efficient influence function
R	Indicator that the full data are observed
π	Observation or response probability

Add chapter-specific notation to this table as the manuscript develops.

Part I

I. Semiparametric foundations

2 Introduction to semiparametric models

2.1 Learning objectives

- Distinguish parametric, nonparametric, and semiparametric models
- Identify parameters of interest and nuisance parameters
- Recognize restricted moment and proportional hazards models as semiparametric

2.2 Prerequisites and notation

2.3 Expository notes

2.4 Expanded derivations

2.5 Worked examples

2.6 Additional exercises

2.7 Chapter summary

3 Hilbert space for random vectors

3.1 Learning objectives

- Work with mean-zero, square-integrable random functions
- Interpret orthogonality through covariance
- Use the projection theorem in statistical problems

3.2 Prerequisites and notation

3.3 Expository notes

3.4 Expanded derivations

3.5 Worked examples

3.6 Solutions to exercises

3.6.1 Exercise 2.1

Solution. **Key idea.**

Derivation.

Result and interpretation.

3.7 Additional exercises

3.8 Chapter summary

4 The geometry of influence functions

4.1 Learning objectives

- Define regular asymptotically linear estimators
- Derive influence functions for M-estimators
- Characterize the efficient influence function geometrically

4.2 Prerequisites and notation

4.3 Expository notes

4.4 Expanded derivations

4.5 Worked examples

4.6 Solutions to exercises

4.7 Additional exercises

4.8 Chapter summary

5 Semiparametric models

5.1 Learning objectives

- Construct parametric submodels
- Define nuisance tangent spaces
- Derive influence functions and efficiency bounds in semiparametric models

5.2 Prerequisites and notation

5.3 Expository notes

5.4 Expanded derivations

5.5 Worked examples

5.6 Solutions to exercises

5.7 Additional exercises

5.8 Chapter summary

6 Other examples of semiparametric models

6.1 Learning objectives

- Analyze location-shift regression models
- Connect proportional hazards estimation to tangent-space geometry
- Derive efficient estimators for means and treatment contrasts

6.2 Prerequisites and notation

6.3 Expository notes

6.4 Expanded derivations

6.5 Worked examples

6.6 Solutions to exercises

6.7 Additional exercises

6.8 Chapter summary

Part II

II. Missing and coarsened data

7 Models and methods for missing data

7.1 Learning objectives

- Distinguish MCAR, MAR, and nonignorable missingness
- Compare likelihood, imputation, and inverse-probability methods
- Explain the basis of doubly robust estimation

7.2 Prerequisites and notation

7.3 Expository notes

7.4 Expanded derivations

7.5 Worked examples

7.6 Solutions to exercises

7.7 Additional exercises

7.8 Chapter summary

8 Missing and coarsening at random for semiparametric models

8.1 Learning objectives

- Formalize missing and coarsened data
- State coarsening-at-random assumptions
- Relate full-data and observed-data models

8.2 Prerequisites and notation

8.3 Expository notes

8.4 Expanded derivations

8.5 Worked examples

8.6 Solutions to exercises

8.7 Additional exercises

8.8 Chapter summary

9 Inverse probability weighted complete-case estimators

9.1 Learning objectives

- Construct IPW complete-case estimating equations
- Establish consistency and asymptotic normality
- Estimate asymptotic variance when response probabilities are estimated

9.2 Prerequisites and notation

9.3 Expository notes

9.4 Expanded derivations

9.5 Worked examples

9.6 Solutions to exercises

9.7 Additional exercises

9.8 Chapter summary

10 Semiparametric estimators with monotone coarsening

10.1 Learning objectives

- Represent monotone coarsening processes
- Derive semiparametric estimators under monotone missingness
- Connect censoring to monotone coarsening

10.2 Prerequisites and notation

10.3 Expository notes

10.4 Expanded derivations

10.5 Worked examples

10.6 Solutions to exercises

10.7 Additional exercises

10.8 Chapter summary

Part III

III. Efficiency and robustness

11 Improving efficiency and double robustness with coarsened data

11.1 Learning objectives

- Project influence functions onto augmentation spaces
- Construct efficient augmented IPW estimators
- Explain double robustness under different coarsening patterns

11.2 Prerequisites and notation

11.3 Expository notes

11.4 Expanded derivations

11.5 Worked examples

11.6 Solutions to exercises

11.7 Additional exercises

11.8 Chapter summary

12 Locally efficient estimators for coarsened-data semiparametric models

12.1 Learning objectives

- Derive observed-data efficient scores
- Compare likelihood and augmented-IPW representations
- Construct locally efficient estimators

12.2 Prerequisites and notation

12.3 Expository notes

12.4 Expanded derivations

12.5 Worked examples

12.6 Solutions to exercises

12.7 Additional exercises

12.8 Chapter summary

13 Approximate methods for gaining efficiency

13.1 Learning objectives

- Define restricted classes of augmented estimators
- Derive optimal estimators within a restricted class
- Balance efficiency gains against computational feasibility

13.2 Prerequisites and notation

13.3 Expository notes

13.4 Expanded derivations

13.5 Worked examples

13.6 Solutions to exercises

13.7 Additional exercises

13.8 Chapter summary

Part IV

IV. Causal inference and imputation

14 Doubly robust estimation of average causal treatment effects

14.1 Learning objectives

- State identification assumptions for point-treatment effects
- Express causal estimation as a coarsened-data problem
- Derive the augmented inverse-probability estimator for the ATE

14.2 Prerequisites and notation

14.3 Expository notes

14.4 Expanded derivations

14.5 Worked examples

14.6 Solutions to exercises

14.7 Additional exercises

14.8 Chapter summary

15 Multiple imputation from a frequentist perspective

15.1 Learning objectives

- Compare full-data and observed-data information
- Derive the asymptotic distribution of multiple-imputation estimators
- Explain proper imputation and Rubin's variance estimator

15.2 Prerequisites and notation

15.3 Expository notes

15.4 Expanded derivations

15.5 Worked examples

15.6 Additional exercises

15.7 Chapter summary

References

Tsiatis, Anastasios A. (2006). Semiparametric theory and missing data. Springer. https://doi.org/10.1007/0-387-37345-4.

A Linear algebra and projections

This appendix will collect matrix identities, projection formulas, and block-matrix calculations used throughout the book.

A.1 Matrix differentiation

A.2 Block-matrix inversion

A.3 Orthogonal projections

B Probability and asymptotic tools

This appendix will collect conditional-expectation identities, convergence results, empirical-process notation, and asymptotic linearity arguments.

B.1 Conditional expectation

B.2 Modes of convergence

B.3 Central limit arguments

C R code and reproducibility

This appendix contains reusable R functions and reproducibility conventions for computational examples.

C.1 Session information

```r
sessionInfo()
```

```
R version 4.5.3 (2026-03-11 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 10 x64 (build 19045)

Matrix products: default
  LAPACK version 3.12.1

locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8

time zone: America/Chicago
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

loaded via a namespace (and not attached):
 [1] compiler_4.5.3    fastmap_1.2.0     cli_3.6.5         htmltools_0.5.9
 [5] tools_4.5.3       otel_0.2.0        rstudioapi_0.18.0 yaml_2.3.12
 [9] tinytex_0.58      rmarkdown_2.30    knitr_1.51        jsonlite_2.0.0
[13] xfun_0.56         digest_0.6.39     rlang_1.1.7       evaluate_1.0.5
```